

PROTOCOL: Sort and array specimens

Insecta v.1.0

This protocol is designed for bulk samples containing both large (pinned) and small (microplated) arthropods preserved in alcohol (e.g., Malaise traps). If your specimens are either or both, you can just refer to the relevant parts of the protocol.

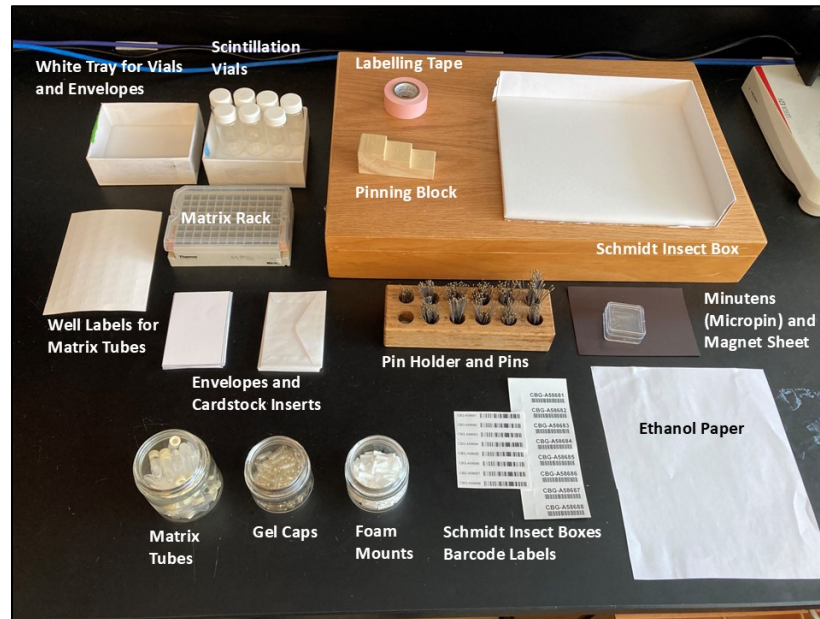
♂ Bullet ♂ points show tips for executing the protocol.

(1) Sort specimens by size

- ♂ Specimen preparation is much more efficient if similarly-sized specimens are processed on the same plates
 - ♂ If all your specimens have the same preservation method (e.g., all are pinned, or all are plated), you can skip to (2)
 - ♂ Non-arthropods should be removed from the workflow at this stage and preserved separately
- a. If your work will involve specimens in microplates, assemble a 'plating' workstation containing as many of the items below as you have. (Labels are printed by ONTOLOGY™). The ONTOLOGY™ also contains a printout for the Voucher Size Guides.



- b. If your work will involve specimens that are too large for microplates, assemble a 'plating' workstation containing as many of the items below as you have. (Labels are printed by ONTOLOGY™).



- c. Sort your specimens into two 'piles' (i) small and (ii) large
- Prepare a 'destination' Petri dish with Ethanol,
 - Using forceps, move large specimens—those that do not fully fit inside the box in the voucher size guide—into the destination Petri dish. When complete, you should have two size-sorted Petri dishes.

(2) Place specimens into 96-well plate format

- ✎ *The later steps of this process will be most efficient if all specimens analyzed on an ONTOLOGY™ plate are arrayed in the same manner—e.g., all pinned or all plated.*
- ✎ *If a 'mixed' plate is necessary, process all of one kind then all of the other, so leave consecutive wells or cells empty.*
- ✎ *You can only analyze $95 \times N$ specimens, where N is then number of ONTOLOGY™ plates remaining in your batch (e.g., 950 specimens if all 10 plates remain). Therefore, if you have more than this number of specimens, choose carefully which you take into the next steps*
- ✎ *Remember that H12 is always empty*

a. Process small specimens into microplates

- Label the plate with the corresponding ONTOLOGY plate ID.
- Pre-fill all wells with EtOH if you will extract DNA at a later date, or with 50 μ L Alkaline Lysis buffer if you will extract DNA immediately (see Protocol – DNA Extraction by Alkaline Lysis)
- Using a microscope to aid you, gently pick up individual specimens with forceps and transfer them to your microplate, starting with well A01 and



proceeding to A12; then rows B–H, ensuring you leave H12 empty except for buffer or preservative.

- iv. After all wells are filled, seal the plate with a thermocycler (PCR) seal
- v. Proceed to the next plate

b. Pin large specimens

- i. For large specimens, pin and place in a labelled pinning box to the positions they will occupy on the plate



- ii. Ensure the box is clearly labelled with the destination ONTOLOGY plate
- iii. Proceed to the next plate

